

MATH 180 Strategies of Problem Solving

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September 27, 2026

Outline

- 1 SoPS Lecture 10
 - Degree sum formula
 - Euler's Formula
 - Wrapping up

Review

Last time

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- we played Sprouts
- intro to graphs
- questions about Sprouts

Outline

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 - Degree sum formula
 - Euler's Formula
 - Wrapping up

Degrees of vertices

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Definition

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Activity

Draw some examples of graphs and for each graph, do the following:

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- Count the vertices

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- Count the edges
- Figure out the degrees of the vertices

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Activity

Draw some examples of graphs and for each graph, do the following:

- Count the vertices
- Count the edges
- Figure out the degrees of the vertices
- Can you spot any relationships?

Degree Sum Formula

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Theorem:

If we add up the degrees of all the vertices in the graph, the result is equal to twice the number of edges.

Degree Sum Formula

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Proof:

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The degree of a vertex counts the number of edges coming out of it.

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The degree of a vertex counts the number of edges coming out of it.

The sum of all the degrees is the number of times any edge comes out of any vertex.

Each edge comes out of exactly two vertices.

Therefore the number of edges is twice the degree sum.

Handshaking Lemma

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Theorem:

The number of vertices which have an odd degree must be even.

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Folksy version:

At any business meeting, different people are shaking hands. The number of people who shook an odd number of other people's hands must be even.

Handshaking Lemma

Proof:

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The sum of the degrees must be even.

If we add up a bunch of numbers

$$a_1 + a_2 + \cdots + a_n,$$

the result is even if and only if an even amount of the a_j 's were odd.

Handshaking Lemma

Proof:

The sum of the degrees must be even.

If we add up a bunch of numbers

$$a_1 + a_2 + \cdots + a_n,$$

the result is even if and only if an even amount of the a_j 's were odd.

Therefore the number of odd-degree vertices must be even.

Outline

1

SoPS Lecture 10

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Connected graphs

Connected graphs

definition

A graph is called **connected** if you can travel from every vertex to every other by walking along a sequence of edges.

definition

More generally, graphs are made up of **connected components**, ie. connected subgraphs which are as large as possible.

Planar graphs

Planar graphs

definition

a graph is called **planar** if none of the edges of the graph overlap.

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activity

draw some examples of connected planar graphs and for each graph, do the following:

- count the vertices
- count the edges
- count the number of regions
- can you spot any relationships?

Euler's formula

Euler's formula

Theorem:

Let E be the number of edges, V the number of faces, and F the number of divided regions of a connected planar graph (including the “outside”). Then

$$V - E + F = 2.$$

Euler's formula

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Proof:

Start with a graph.

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If it has an enclosed region, snip a boundary edge, creating a new graph.

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Start with a graph.

If it has an enclosed region, snip a boundary edge, creating a new graph.

This merges the the region with an adjacent region, keeping the graph connected.

Therefore the new graph has one less edge, one less region, but the same number of vertices.

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Start with a graph.

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Therefore it doesn't change the value of $V - E + F$.

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Start with a graph.

If it has an enclosed region, snip a boundary edge, creating a new graph.

This merges the the region with an adjacent region, keeping the graph connected.

Therefore the new graph has one less edge, one less region, but the same number of vertices.

Therefore it doesn't change the value of $V - E + F$.

Continue in this way until there's no enclosed regions, so $F = 1$.

Euler's formula

Euler's formula

Proof (continued)

Euler's formula

Proof (continued)

When $F = 1$, the graph is a **tree**, connected without loops.

Euler's formula

Proof (continued)

When $F = 1$, the graph is a **tree**, connected without loops. It will have an vertex of degree 1 at the tip of a branch.

Euler's formula

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When $F = 1$, the graph is a **tree**, connected without loops. It will have an vertex of degree 1 at the tip of a branch. Remove that vertex and the edge it touched, "pruning" the tree.

Euler's formula

Proof (continued)

When $F = 1$, the graph is a **tree**, connected without loops. It will have an vertex of degree 1 at the tip of a branch. Remove that vertex and the edge it touched, "pruning" the tree. This removes one edge and one vertex, leaving a smaller tree without changing the value of $V - E + F$.

Euler's formula

Proof (continued)

When $F = 1$, the graph is a **tree**, connected without loops. It will have an vertex of degree 1 at the tip of a branch. Remove that vertex and the edge it touched, "pruning" the tree. This removes one edge and one vertex, leaving a smaller tree without changing the value of $V - E + F$. Prune until you are left with a single vertex and edge.

Euler's formula

Proof (continued)

When $F = 1$, the graph is a **tree**, connected without loops.

It will have an vertex of degree 1 at the tip of a branch.

Remove that vertex and the edge it touched, "pruning" the tree.

This removes one edge and one vertex, leaving a smaller tree without changing the value of $V - E + F$.

Prune until you are left with a single vertex and edge.

You never changed the value of $V - E + F$, so $V - E + F = 2$.

Multi-component Euler's formula

Multi-component Euler's formula

Theorem:

Let E be the number of edges, V the number of faces, and F the number of divided regions and C the number of connected components of a planar graph (including the "outside"). Then

$$V - E + F = 1 + C.$$

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Question of the Day

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Puzzle

We say a graph can be painted with 3 colors if you can paint its edges red, blue, and green such that no vertex has multiple edges of the same color coming out.

Find an example of a graph where every vertex has degree 3, but where the graph is not 3-colorable.

Final thoughts

Reflection

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Reflection

- What does it mean for a graph to be connected?

Final thoughts

Reflection

- What does it mean for a graph to be connected?
- What does it mean for a graph to be a tree?

Submit your responses by **Tonight!**